Notes: Ljungqvist, Marston, Starks, Wei & Yan (JFE, 2007)

Conflicts of Interest in Sell-Side Research and the Moderating Role of Institutional Investors

Contents

1 Big picture

- **Question.** Do institutional investors discipline sell-side analysts when analysts face investment banking and brokerage conflicts of interest?
- **Core idea.** Analysts depend on institutional investors for career rewards, all-star votes, and trading commission allocations. Therefore, analysts should be less willing to issue biased, overly favorable research in stocks that are highly visible to institutions.
- **Main setting.** The paper studies a large panel of analyst recommendations from I/B/E/S over 1994–2000, merged with institutional ownership data from Thomson 13f, underwriting relationships from SDC, bank characteristics, brokerage size, analyst reputation, and forecast accuracy.
- **Main dependent variable.** The main outcome is the analyst's recommendation for a stock relative to the consensus recommendation for that stock. Recommendations are coded from 1 for sell to 5 for strong buy; the relative recommendation is the analyst's recommendation minus the median recommendation for the same firm.
- **Investment banking conflict.** Analysts are more optimistic toward firms with which their bank has stronger prior underwriting relationships, especially equity relationships.
- **Brokerage conflict.** Analysts employed by banks with larger brokerage operations issue more favorable relative recommendations, consistent with pressure to stimulate trading commissions.
- **Institutional monitoring result.** Recommendations are less optimistic when the stock has higher institutional ownership. Institutional ownership moderates conflicts by raising the reputational cost of biased research.
- **Forecast accuracy result.** Analysts produce more accurate earnings forecasts in stocks with higher institutional ownership. This implies institutional investors do not merely discipline recommendation levels; they also affect research quality.
- **Bad-news reaction result.** Analysts react more quickly to large negative stock price shocks when institutional ownership is high. Institutional ownership therefore also disciplines timeliness.
- **Central takeaway.** Sell-side analyst conflicts are real, but their effects are weaker when sophisticated institutional investors are present and able to reward or punish analyst behavior.

2 Incremental contribution of this paper

2.1 Relative to the investment-banking conflict literature

Earlier studies showed that analysts affiliated with underwriters can be overly optimistic around IPOs, seasoned equity offerings, or other underwriting events. This paper contributes by moving beyond narrow event windows and studying a broad cross-section of analyst recommendations for firms regardless of whether they are currently issuing securities.

- It treats investment banking pressure as a continuing relationship variable rather than a one-time event dummy.
- It measures the strength of the bank-company relationship using the bank's share of the firm's prior equity and debt underwriting mandates.
- It shows that the optimism associated with bank relationships is present in general analyst coverage, not only immediately around securities issuance.

Interpretation: The contribution is to show that conflicts of interest are embedded in ongoing bank-client relationships, not only in high-profile issuance events.

2.2 Relative to the brokerage-pressure literature

Other studies argue that analysts have incentives to publish bullish research because optimistic recommendations stimulate trading volume. This paper adds a bank-level brokerage-pressure proxy: the number of registered representatives, which captures the size of the bank's brokerage business.

- Analysts at banks with larger brokerage operations issue more favorable relative recommendations.
- This result is estimated alongside investment-banking relationship variables, so the paper can separate brokerage pressure from underwriting pressure.

Interpretation: The paper's incremental contribution is to combine the two main conflicts–investment banking and brokerage–in one unified empirical design.

2.3 Relative to research on analyst reputation

Prior work emphasizes analyst reputation, such as Institutional Investor all-star status. This paper argues that the ultimate source of much analyst reputation comes from institutional investors themselves.

- Institutional investors vote in analyst rankings and influence analyst compensation and career prospects.
- Institutions allocate trading commissions across brokerage houses partly based on the perceived quality of research.
- Therefore, the presence of institutional investors in a stock should directly change analysts' incentives.

Interpretation: Instead of only studying whether high-reputation analysts behave differently, the paper studies the market participants who create and enforce much of that reputation.

2.4 Relative to studies of institutional investor monitoring

Institutional investor monitoring is often studied through governance, activism, or block ownership. This paper extends that literature to the information intermediation role of sell-side analysts.

- Institutional ownership disciplines not only managers but also financial intermediaries.
- Institutional monitoring changes analyst recommendations, forecast accuracy, and reaction speed to bad news.
- The disciplinary effect appears even though institutions do not directly employ the sell-side analyst.

Interpretation: The paper broadens the meaning of institutional investor monitoring: institutions influence the quality of information production in capital markets.

2.5 Relative to earnings-forecast studies

Analyst earnings forecasts can be optimistic or pessimistic depending on incentives to help firms beat forecasts. Recommendations are less subject to this competing pessimism bias. The paper therefore focuses on relative recommendations as its main outcome while using forecast accuracy as a separate quality test.

Interpretation: The paper's recommendation-based design helps avoid the interpretive problem that optimistic and pessimistic forecast incentives may offset each other.

3 Data and measurement

3.1 Sample construction

- The sample period is 1994–2000.
- Recommendations come from Thomson I/B/E/S.
- Institutional ownership comes from Thomson 13f filings.
- Underwriting relationships come from Thomson SDC New Issues.
- Bank brokerage size comes from the Securities Industry Yearbook.
- Analyst characteristics include all-star status, forecast accuracy history, seniority, job moves, coverage initiation, seasoning, and number of stocks covered.
- The main recommendations sample contains 230,268 firm-analyst-quarter observations after excluding anonymous recommendations.

Interpretation: The empirical design is unusually rich because it observes analyst, bank, firm, relationship, and institutional ownership variables in the same panel.

3.2 Analyst recommendations

Recommendations are coded as:

$$1 = \text{sell}, \quad 2 = \text{underperform}, \quad 3 = \text{hold}, \quad 4 = \text{buy}, \quad 5 = \text{strong buy}. \tag{1}$$

The main outcome is the relative recommendation:

$$A_{i,k,t} = Recommendation_{i,k,t} - MedianRecommendation_{k,t}. \tag{2}$$

- i indexes analyst.
- k indexes firm.
- t indexes quarter.
- A positive value means the analyst is more bullish than the consensus for that stock.

Interpretation: This measure controls for differences in average optimism across stocks. It asks whether a given analyst is more favorable than other analysts covering the same firm.

3.3 Institutional ownership

The main institutional ownership measure is:

$$InstOwn_{k,t} = \frac{\text{shares held by 13f institutions}}{\text{shares outstanding}}. \tag{3}$$

The paper also studies:

- the log number of institutional owners;
- the mean size of institutional holdings;
- ownership by the top 100 largest institutions;
- ownership by smaller institutions.

Interpretation: Institutional ownership is both a demand-for-information variable and a monitoring/visibility variable. More institutional ownership means analysts face greater scrutiny.

3.4 Investment banking pressure

Investment banking pressure is proxied by the strength of the prior underwriting relationship between the analyst's bank and the covered firm.

For bank j and firm k, the relationship measures are:

$$EquityRel_{j,k,t} = \frac{\text{equity proceeds lead-managed by bank } j \text{ for firm } k \text{ in prior five years}}{\text{total equity proceeds raised by firm } k \text{ in prior five years}}, \tag{4}$$

$$DebtRel_{j,k,t} = \frac{\text{debt proceeds lead-managed by bank } j \text{ for firm } k \text{ in prior five years}}{\text{total debt proceeds raised by firm } k \text{ in prior five years}}. \tag{5}$$

Interpretation: A stronger bank-company relationship means the analyst's employer has more to gain from pleasing the firm and preserving future investment banking business.

3.5 Brokerage pressure

Brokerage pressure is proxied by:

$$BrokerageSize_{j,t} = \log(\text{number of registered representatives at bank } j). \tag{6}$$

Interpretation: If bullish recommendations stimulate trading, then analysts at banks with larger brokerage operations have stronger incentives to be optimistic.

3.6 Bank characteristics

The paper uses two key bank characteristics:

- **Equity underwriting market share:** a proxy for bank reputation capital.
- **Loyalty index:** the fraction of equity clients retained across consecutive deals.

The loyalty index is conceptually:

$$Loyalty_j = \frac{\text{number of clients retained by bank } j}{\text{number of clients previously served by bank } j}. \tag{7}$$

Interpretation: More reputable or relationship-focused banks may be less willing to damage long-term reputation by encouraging visibly biased research.

3.7 Analyst characteristics

The analyst-level controls include:

- Institutional Investor all-star status;
- analyst's historical relative forecast accuracy;
- seniority;
- coverage initiation indicator;
- seasoning since coverage began;
- job move indicator;
- number of stocks covered.

Interpretation: These variables control for differences in analyst ability, reputation, career stage, and workload.

3.8 Forecast accuracy

Forecast accuracy is measured relative to consensus. The paper first computes scaled absolute forecast error:

$$Error_{i,k,t} = \frac{|RealizedEarnings_k - Forecast_{i,k,t}|}{SD(Forecasts_{k,t})}. \tag{8}$$

It then subtracts the scaled absolute error of the consensus forecast:

$$Accuracy_{i,k,t} = Error_{i,k,t} - Error_{Consensus,k,t}. \tag{9}$$

Interpretation: Lower values mean the analyst is more accurate relative to the consensus. Negative coefficients in the forecast accuracy regressions therefore indicate smaller errors and better accuracy.

4 Empirical specifications

4.1 Main recommendation model

The main model is:

$$A_{i,k,t} = \alpha C_{k,t} + \beta I_{i,t} + \theta R^j_{i,k,t} + \gamma B^j_{i,t} + \epsilon_{i,k,t}. \tag{10}$$

- $A_{i,k,t}$ is analyst i's relative recommendation for firm k in quarter t.
- $C_{k,t}$ contains firm characteristics, institutional ownership, and fee-generating capacity.
- $I_{i,t}$ contains analyst characteristics.
- $R^j_{i,k,t}$ contains bank-company relationship variables.
- $B^j_{i,t}$ contains bank characteristics and brokerage pressure.

Interpretation: The model compares recommendations across analysts, firms, and time while controlling for many sources of incentives.

4.2 Error decomposition

The error is decomposed as:

$$\epsilon_{i,k,t} = \eta_i + u_k + v_t + \nu_{i,k,t}, \tag{11}$$

where:

- η_i captures analyst-specific unobservables;
- u_k captures firm-specific unobservables;
- v_t captures time effects;
- $\nu_{i,k,t}$ is the residual.

Interpretation: Because the panel varies along analyst, firm, and time dimensions, the paper considers specifications with analyst random effects, firm random effects, and Fama-MacBeth methods.

4.3 Forecast accuracy model

The forecast accuracy model uses the same right-hand-side variables as the recommendation model:

$$Accuracy_{i,k,t} = \alpha C_{k,t} + \beta I_{i,t} + \theta R^j_{i,k,t} + \gamma B^j_{i,t} + \epsilon_{i,k,t}. \tag{12}$$

Interpretation: This model asks whether conflicts and institutional ownership affect research quality, not just recommendation optimism.

4.4 Bad-news reaction-time model

To study timeliness after large price drops, the paper estimates:

$$\ln T_{i,k} = \pi \Delta P_k + \kappa C_k + \lambda I_i + \omega R^j_{i,k} + \phi B^j_i + \mu_{i,k}. \tag{13}$$

- $T_{i,k}$ is the time until analyst i issues a new recommendation after a large one-day price drop in firm k.
- ΔP_k is the one-day stock price change.
- Large drops are defined as 4 or 5 times the firm's prior-year daily return standard deviation.

Interpretation: If analysts delay bad news for relationship clients, relationship variables should increase reaction time. If institutions discipline analysts, institutional ownership should reduce reaction time.

5 Identification and empirical design

5.1 Main identifying variation

The identification comes from three sources:

1. different analysts cover the same firm at the same time;
2. the same analyst covers different firms with different institutional ownership and banking relationships;
3. banks differ in brokerage size, reputation, loyalty, and relationships with covered firms.

Interpretation: The panel structure allows the authors to separate analyst incentives, firm characteristics, bank characteristics, and bank-firm relationships.

5.2 Why relative recommendations are useful

Using recommendations relative to consensus removes stock-level optimism and pessimism common to all analysts covering a firm.

- A stock may be genuinely attractive and receive bullish recommendations from everyone.
- Relative recommendations ask whether a particular analyst is more bullish than the peer group covering that same stock.
- This is well suited to studying conflicts of interest.

Interpretation: The relative measure makes the outcome closer to analyst-specific bias rather than stock-level attractiveness.

5.3 Addressing coverage selection

A potential concern is endogenous coverage selection: analysts might choose to cover firms where their bank has relationships or where they are predisposed to be optimistic. The paper addresses this in Table 3 by estimating Heckman selection models.

Interpretation: The key coefficients remain similar, suggesting that endogenous coverage choice does not drive the main results.

5.4 Why institutional ownership can discipline analysts

Institutional investors discipline analysts through several channels:

- voting in Institutional Investor all-star rankings;
- allocating trading commissions across brokerage firms;
- rewarding analysts who provide accurate, timely, and useful research;
- punishing analysts whose research appears biased.

Interpretation: Institutional ownership is not merely a control variable. It is the main mechanism that raises the analyst's cost of publishing biased research.

5.5 Limitations

- Institutional ownership is not randomly assigned.
- The model cannot observe the analyst's compensation contract directly.
- The brokerage-pressure proxy is imperfect because it uses registered representatives rather than brokerage revenue.
- Recommendation bias, forecast accuracy, and reaction time are related but not identical dimensions of research quality.
- The sample is pre-Regulation AC and pre-Global Settlement, so the institutional environment differs from later sell-side research markets.

Interpretation: The paper's strength is not experimental randomization; it is the detailed simultaneous measurement of institutional ownership, bank relationships, brokerage pressure, and analyst behavior.

6 Tables

6.1 Table 1: Summary statistics

Read:

- The sample contains 230,268 firm-analyst-quarter recommendation observations.
- The average firm is covered by 5.6 analysts.
- Average institutional ownership is 52.8%, with a median of 55.5%.
- Banks' average equity relationship share with covered companies is 10.7%, while the average debt relationship share is 4.1%.
- 29.2% of recommendations are issued by Institutional Investor all-star analysts.
- The average relative recommendation is close to zero, as expected after subtracting stock-level consensus.

Economic message: The sample is large and rich enough to study analyst, bank, firm, relationship, and institutional-owner incentives jointly.

Table 1
Descriptive sample statistics

The sample consists of 230,268 firm-analyst quarters, representing the intersection of the Thomson 13f and Institutional Brokers' Estimates System (I/B/E/S) analyst recommendations data sets during the period 1994–2000. We exclude companies in Standard Industrial Classification industries 6000–6999 (financial institutions, etc.) and 9000–9999 (government agencies, etc.), and 5,264 cases in which analysts submit recommendations to I/B/E/S anonymously.

Variable	Number	Mean	Standard deviation	Minimum	Median	Maximum
Number of analysts covering stock	230,268	5.6	3.9	1	5	36
Company's institutional ownership (percent)	230,268	52.8	22	0	55.5	100
Company's equity market capitalization (millions of dollars)	229,990	6,645	21,900	0	1,218	602,000
Company's equity proceeds prior five years (millions of dollars)	131,399	320.7	1,102.1	0.2	114.1	67,064
Company's debt proceeds prior five years (millions of dollars)	103,805	936.5	1,759.9	0.2	330	43,633
Bank's equity market share prior calendar year (percent)	230,268	5.2	5	0	3.3	21.5
Bank's loyalty index (percent of retained clients)	230,268	62.2	16.8	0	64.6	100
Number of registered representatives (i.e., brokerage employees)	230,268	3,798	5,039	20	948	19,000
Bank's share of company's equity deals prior five years (percent)	230,268	10.7	29.9	0	0	100
Bank's share of company's debt deals prior five years (percent)	230,268	4.1	17.4	0	0	100
Institutional Investor all stars (percent of recommendations)	230,268	29.2				
Analyst's relative forecast accuracy ($\times$ 100)	224,669	51.8	10.2	0	52.4	100
Analyst's seniority (in years)	230,268	6.9	4.8	0	6	19
Jobmove (percent of analysts changing employer)	230,268	1.5				
Relative recommendations	230,268	0.015	0.729	−4	0	4
Initiation (percent of first-time recommendations)	230,268	9.7				
Number of quarters since analyst initiated coverage (seasoning)	230,268	9.9	12.2	0	5	75
Number of stocks covered by analyst	230,268	13.0	11.2	1	11	115
Forecast accuracy (in standard deviations relative to consensus)	238,418	0.030	0.818	−4.2	0	4.9

6.2 Table 2: Determinants of relative recommendations

Read:

- Equity and debt underwriting relationships are positively related to relative recommendations.
- Brokerage pressure is positive: analysts at banks with more registered representatives issue more favorable recommendations.
- Bank loyalty and bank equity market share are generally associated with more conservative recommendations.
- Institutional ownership has a negative coefficient across specifications, meaning analysts are less optimistic when institutional ownership is high.
- Analyst reputation and ability variables matter: all-star status and job moves are associated with less aggressive recommendations in some specifications, while relative forecast accuracy is positively associated with recommendations in the analyst-effects model.

Economic message: Investment banking and brokerage conflicts push recommendations upward, while institutional ownership and reputation-related bank characteristics restrain optimism.

Table 2
Determinants of relative recommendations

The dependent variable is relative recommendations, defined as analyst i's recommendation for company k's stock relative to consensus (the median recommendation of all other analysts covering firm k), where recommendations are coded from 1 (sell) to 5 (strong buy). By construction, relative recommendations lie between −4 and +4 with positive values denoting relatively aggressive recommendations. The models in Columns 1 and 3 are estimated using random-effects generalized least squares (GLS) regressions, including either analyst or firm random effects, respectively. The last three lines of the table provide statistical tests that are only appropriate for the random effects GLS models in Columns 1 and 3. The Fama and MacBeth estimates in Column 2 are based on the time-series average coefficients from 28 separate regressions (one for each possible quarter). The standard errors are obtained from the time-series variation of the coefficient estimates. The ordered probit in the last column models a three-level choice: issuing a below-, at- or above-consensus recommendation. (Ordered probits cannot accommodate fixed or random effects.) The ordered probit standard errors are clustered by analyst (i.e., observations are assumed to be independent across analysts but not necessarily within). To conserve space, intercepts effects are not shown. Standard errors are shown in italics underneath the coefficient estimates. We use ***, **, and * to denote significance at the 0.1%, 1%, and 5% level (two-sided), respectively. The number of observations in all models is 224,404. This is 5,864 short of the available number of firm-analyst quarters, because of missing observations on relative forecast accuracy, analyst seniority, and equity market capitalization.

	Dependent variable: relative recommendations			
Independent variables	Analyst random-effects (1)	Fama and MacBeth (2)	Firm random-effects (3)	Ordered probit (4)
Investment banking pressure (bank-company relationships)				
Bank's share of company's equity deals prior five years	0.094***	0.094***	0.091***	0.135***
	0.006	*0.005*	*0.006*	*0.016*
Bank's share of company's debt deals prior five years	0.106***	0.102***	0.141***	0.201***
	0.009	*0.009*	*0.009*	*0.026*
Brokerage pressure (size of brokerage)				
Log number of registered representatives	0.012***		0.024***	0.037***
	0.002		*0.001*	*0.005*
Bank characteristics				
Bank's loyalty index	−0.109***		−0.174***	−0.273***
	0.013		*0.011*	*0.048*
Bank's equity market share prior calendar year	−0.448***		0.010	0.059
	0.059		*0.034*	*0.168*
Institutional ownership				
Percent institutional ownership	−0.072***	−0.066***	−0.087***	−0.089***
	0.008	*0.010*	*0.010*	*0.027*

Table 2, part 1

Table 2 (*continued*)

	Dependent variable: relative recommendations			
Independent variables	Analyst random-effects (1)	Fama and MacBeth (2)	Firm random-effects (3)	Ordered probit (4)
Analyst characteristics				
Equals one if analyst is an *Institutional Investor* "all-star"	0.003		−0.026***	−0.028
	0.006		*0.004*	*0.024*
Relative forecast accuracy	0.079***		0.003	0.003
	0.022		*0.015*	*0.076*
Log analyst's seniority (in years)	0.001		0.018***	0.022
	0.005		*0.003*	*0.013*
Equals one if analyst initiates coverage	0.000	0.000	0.009	0.015
	0.006	*0.007*	*0.006*	*0.011*
Log number of quarters since analyst initiated coverage (seasoning)	0.019***	0.018***	0.021***	0.037***
	0.002	*0.002*	*0.002*	*0.007*
Jobmove (analyst changing employer)	−0.040***		−0.061***	−0.086*
	0.011		*0.010*	*0.034*
Log number of stocks covered by analyst	−0.043***		−0.037***	−0.047*
	0.004		*0.003*	*0.020*
Company characteristics				
Log number of analysts covering stock	0.088***	0.084***	0.079***	0.122***
	0.004	*0.006*	*0.004*	*0.012*
Log company's dollar equity proceeds prior five years	−0.007***	−0.007***	−0.006***	−0.009***
	0.001	*0.001*	*0.001*	*0.002*
Log company's dollar debt proceeds prior five years	0.0019**	0.0019*	0.0016	0.0012
	0.0006	*0.0008*	*0.0009*	*0.0022*
Log company's dollar equity market capitalization	0.002	0.003	0.002	0.006
	0.001	*0.002*	*0.002*	*0.004*
Random analyst effects	Yes	Yes	No	No
Random firm effects	No	No	Yes	No
Fraction of variance resulting from random effects (ρ) (percent)	14.7	—	3.3	—
Wald test: all coefficients = 0 (F)	118.5***	—	146.4***	32.0***
Overall GLS R^2/pseudo R^2 (percent)	1.2	—	1.4	0.7

Table 2, part 2

6.3 Table 3: Endogenous coverage selection

Read:

- The table checks whether coverage selection drives the main recommendation results.
- Columns for large firms replicate the baseline recommendation results in a subsample where coverage discretion is reduced.
- The Heckman model explicitly estimates selection into coverage.
- The equity and debt relationship coefficients remain positive.
- Institutional ownership remains negatively related to relative recommendations.

Economic message: The main results are not simply caused by analysts choosing which firms to cover. Conflicts and institutional monitoring appear within covered stocks.

The models in Columns 1 and 2 replicate Models 1 and 3 from Table 2 for the subsample of large firms, defined as the five largest firms in each three-digit standard industrial classification code, ranked quarterly by sales. Analysts arguably have less discretion with respect to covering the largest companies. Models 1 and 2 are estimated using random-effects GLS regressions, including either analyst or firm random effects, respectively. Model 3 uses the full sample and is estimated as a Heckman (1979) selection model. The first step models whether a given analyst i covers a given stock k. This is the case in 5.6% of firm-analyst quarters. To instrument the choice, we include the fraction of firms in company k's Fama and French (1997) industry that analysts at i's bank cover at time t. The broader the bank's existing coverage of an industry, the lower the cost of covering company k's stock. This variable is uncorrelated with the second-step residuals. The second step of Model 3 is estimated using the Maximum Likelihood Estimation version of Heckman (1979). To conserve space, intercepts effects are not shown. We show fraction of variance resulting from random effects and an overall GLS R^2 for the GLS random effects models in Columns 1 and 2. We show the likelihood ratio test for the Heckman model in Column 3. Standard errors are shown in italics underneath the coefficient estimates. We use ***, **, and * to denote significance at the 0.1%, 1%, and 5% level (two-sided), respectively. The number of observations in Models 1 and 2 is 80,813. The number of uncensored and censored observations in Model 3 is 224,404 and 3,769,805, respectively.

	Dependent variable			
	Large firms only		Heckman model	
	Relative recommendations			
Independent variables	Analyst effects (1)	Firm effects (2)	Equals one if analyst covers stock (3)	Relative recommendations
Investment banking pressure (bank-company relationships)				
Bank's share of company's equity deals prior five years	0.120***	0.114***	1.478***	0.183***
	0.013	*0.013*	*0.007*	*0.010*
Bank's share of company's debt deals prior five years	0.114***	0.152***	0.361***	0.157***
	0.015	*0.015*	*0.011*	*0.009*
Brokerage pressure (size of brokerage)				
Log number of registered representatives	0.010***	0.023***	0.047***	0.030***
	0.003	*0.002*	*0.001*	*0.001*
Bank characteristics				
Bank's loyalty index	−0.119***	−0.193***	0.424***	−0.167***
	0.023	*0.019*	*0.008*	*0.011*
Bank's equity market share prior calendar year	−0.401***	0.054	−0.176***	0.141**
	0.098	*0.058*	*0.044*	*0.048*
Institutional ownership				
Percent institutional ownership	−0.089***	−0.120***	0.200***	−0.070***
	0.015	*0.020*	*0.006*	*0.008*
Analyst characteristics				
Equals one if analyst is an *Institutional Investor* "all-star"	0.008	−0.021***		−0.021***
	0.009	*0.006*		*0.004*

Table 3, part 1

Table 3 (*continued*)

	Dependent variable			
	Large firms only		Heckman model	
	Relative recommendations			
Independent variables	Analyst effects (1)	Firm effects (2)	Equals one if analyst covers stock (3)	Relative recommendations
Relative forecast accuracy	0.137***	0.061*		0.004
	0.040	*0.028*		*0.015*
Log analyst's seniority (in years)	−0.013	0.015***		0.011***
	0.008	*0.005*		*0.002*
Equals one if analyst initiates coverage	0.005	0.020		0.015*
	0.012	*0.013*		*0.006*
Log number of quarters since analyst initiated coverage (seasoning)	0.018***	0.021***		0.029***
	0.004	*0.004*		*0.002*
Jobmove (analyst changing employer)	−0.033	−0.068***		−0.052***
	0.019	*0.019*		*0.010*
Log number of stocks covered by analyst	−0.042***	−0.033***		−0.032***
	0.006	*0.005*		*0.003*
Company characteristics				
Log number of analysts covering stock	0.082***	0.077***	1.122***	0.166***
	0.007	*0.007*	*0.003*	*0.008*
Log company's dollar equity proceeds prior five years	−0.007***	−0.006***	−0.010***	−0.008***
	0.001	*0.001*	*0.001*	*0.001*
Log company's dollar debt proceeds prior five years	0.001	0.002	−0.026***	0.0014*
	0.001	*0.001*	*0.001*	*0.0006*
Log company's dollar equity market capitalization	0.005*	0.005	0.019***	0.002
	0.002	*0.003*	*0.001*	*0.001*
Instrument				
Fraction of Fama and French (1997) industry covered by bank			2.185***	
			0.011	
Random analyst effects	Yes	No	No	No
Random firm effects	No	Yes	No	No
Fraction of variance resulting from random effects (ρ) (percent)	17.0	2.0	n.a.	n.a.
Wald test: all coefficients = 0 (F)	37.2***	50.5***	151.7***	
Overall GLS R^2 (percent)	1.0	1.3	n.a.	
Likelihood ratio test: independent equations ($\rho = 0$) (χ^2)	n.a.	n.a.	126.2***	

Table 3, part 2

6.4 Table 4: Number and size of institutional owners

Read:

- Table 4 replaces percentage institutional ownership with the log number of institutional investors and the mean size of institutional holdings.
- The log number of institutional investors is negative in the firm random-effects specification.
- Mean size of institutional holdings is strongly negative across specifications.

- Relationship and brokerage pressure remain positive.

Economic message: Institutional monitoring is not just about total percentage ownership. The visibility and concentration of institutional investors also matter for analyst discipline.

Table 4
The effect of the number of institutional owners

The dependent variable is relative recommendations, defined as analyst i's recommendation for company k's stock relative to consensus (the median recommendation of all other analysts covering firm k). All models are estimated using random-effects generalized least squares (GLS), including either analyst or firm effects. This table replaces institutional investor holdings from Table 2 with two alternative institutional ownership variables: the log number of institutional investors (see Columns 1 and 2), and the mean size of institutional investor holdings, computed as (institutional investor holdings)/(number of institutional investors) (see Columns 3 and 4). To conserve space, intercepts are not shown. Standard errors are shown in italics underneath the coefficient estimates. We use ***, **, and * to denote significance at the 0.1%, 1%, and 5% level (two-sided), respectively. The number of observations is 224,280.

	Dependent variable: relative recommendations			
Independent variables	(1)	(2)	(3)	(4)
Investment banking pressure (bank-company relationships)				
Bank's share of company's equity deals prior five years	0.097***	0.092***	0.097***	0.094***
	0.006	*0.006*	*0.006*	*0.006*
Bank's share of company's debt deals prior five years	0.105***	0.141***	0.105***	0.141***
	0.009	*0.009*	*0.009*	*0.009*
Brokerage pressure (size of brokerage)				
Log number of registered representatives	0.012***	0.024***	0.012***	0.024***
	0.002	*0.001*	*0.002*	*0.001*
Bank characteristics				
Bank's loyalty index	−0.109***	−0.174***	−0.108***	−0.173***
	0.013	*0.011*	*0.013*	*0.011*
Bank's equity market share prior calendar year	−0.452***	0.007	−0.452***	0.005
	0.059	*0.034*	*0.059*	*0.034*
Institutional ownership				
Log number of institutional investors	−0.007	−0.016***		
	0.004	*0.005*		
Mean size of institutional investor holdings			−0.494***	−0.473***
			0.140	*0.153*
Analyst characteristics				
Equals one if analyst is an *Institutional Investor* "all-star"	0.004	−0.026***	0.003	−0.026***
	0.006	*0.004*	*0.006*	*0.004*

Table 4, part 1

Table 4 (*continued*)

	Dependent variable: relative recommendations			
Independent variables	(1)	(2)	(3)	(4)
Relative forecast accuracy	0.079***	0.005	0.078***	0.003
	0.022	*0.015*	*0.022*	*0.015*
Log analyst's seniority (in years)	0.002	0.018***	0.002	0.018***
	0.005	*0.003*	*0.005*	*0.003*
Equals one if analyst initiates coverage	0.000	0.009	−0.001	0.009
	0.006	*0.006*	*0.006*	*0.006*
Log number of quarters since analyst initiated coverage (seasoning)	0.018***	0.021***	0.017***	0.020***
	0.002	*0.002*	*0.002*	*0.002*
Jobmove (analyst changing employer)	−0.041***	−0.061***	−0.041***	−0.061***
	0.011	*0.010*	*0.011*	*0.010*
Log number of stocks covered by analyst	−0.042***	−0.036***	−0.043***	−0.037***
	0.004	*0.003*	*0.004*	*0.003*
Company characteristics				
Log number of analysts covering stock	0.085***	0.079***	0.084***	0.077***
	0.004	*0.004*	*0.004*	*0.004*
Log company's dollar equity proceeds prior five years	−0.007***	−0.006***	−0.007***	−0.006***
	0.001	*0.001*	*0.001*	*0.001*
Log company's dollar debt proceeds prior five years	0.002**	0.002*	0.002**	0.001
	0.0006	*0.0009*	*0.0006*	*0.001*
Log company's dollar equity market capitalization	0.004	0.006	−0.001	−0.003
	0.002	*0.003*	*0.001*	*0.002*
Random analyst effects	Yes	No	Yes	No
Random firm effects	No	Yes	No	Yes
Fraction of variance resulting from random effects (ρ) (percent)	14.8	3.3	14.8	3.3
Wald test: all coefficients = 0 (χ^2)	113.5***	142.5***	114.1***	142.6***
Overall GLS R^2 (percent)	1.2	1.4	1.2	1.4

Table 4, part 2

6.5 Table 5: Composition of institutional ownership

Read:

- Table 5 separates the top 100 largest institutions from smaller institutions.
- Institutional ownership by top 100 investors is negatively related to relative recommendations.
- Ownership by other investors is also negative in some specifications.
- The average holding size of the top 100 institutions has the strongest moderating effect.

Economic message: The disciplinary effect of institutional ownership is strongest when large, visible institutions hold meaningful stakes.

The effect of the composition of institutional ownership

The dependent variable is relative recommendations, defined as analyst i's recommendation for company k's stock relative to consensus (the median recommendation of all other analysts covering firm k). All models are estimated using random-effects generalized least squares (GLS), including either analyst or firm effects. In this table, we distinguish between the one hundred largest institutions judged by their aggregate portfolio valuations as of the last quarter of 1993 and smaller investors. In cols. (1) and (2), we split overall institutional ownership into the holdings of top one hundred investors and the holdings of smaller investors. In cols. (3) and (4) we include the mean size of institutional holdings for these two groups (computed as aggregate ownership/number of investors). To conserve space, intercepts are not shown. Standard errors are shown in italics underneath the coefficient estimates. We use ***, **, and * to denote significance at the 0.1%, 1%, and 5% level (two-sided), respectively. The number of observations is 224,352 in Columns 1 and 2 and 223,721 in Columns 3 and 4.

	Dependent variable: relative recommendations			
Independent variables	(1)	(2)	(3)	(4)
Investment banking pressure (bank-company relationships)				
Bank's share of company's equity deals prior five years	0.094***	0.091***	0.097***	0.093***
	0.006	*0.006*	*0.006*	*0.006*
Bank's share of company's debt deals prior five years	0.105***	0.141***	0.105***	0.141***
	0.009	*0.009*	*0.009*	*0.009*
Brokerage pressure (size of brokerage)				
Log number of registered representatives	0.012***	0.024***	0.012***	0.024***
	0.002	*0.001*	*0.002*	*0.001*
Bank characteristics				
Bank's loyalty index	−0.109***	−0.174***	−0.109***	−0.174***
	0.013	*0.011*	*0.013*	*0.011*
Bank's equity market share prior calendar year	−0.451***	0.010	−0.451***	0.007
	0.059	*0.034*	*0.059*	*0.034*
Institutional ownership				
Percent institutional ownership: top one hundred investors	−0.095***	−0.094***		
	0.013	*0.016*		
Percent institutional ownership: other investors	−0.047***	−0.080***		
	0.013	*0.016*		
Mean size of institutional holdings: top one hundred investors			−0.414**	−0.477**
			0.138	*0.154*
Mean size of institutional holdings: other investors			−0.134	0.090
			0.235	*0.257*

Table 5, part 1

Table 5 (*continued*)

	Dependent variable: relative recommendations			
Independent variables	(1)	(2)	(3)	(4)
Analyst characteristics				
Equals one if analyst is an *Institutional Investor* all-star	0.003	−0.026***	0.004	−0.026***
	0.006	*0.004*	*0.006*	*0.004*
Relative forecast accuracy	0.078***	0.004	0.078***	0.003
	0.022	*0.015*	*0.022*	*0.015*
Log analyst's seniority (in years)	0.000	0.018***	0.002	0.018***
	0.005	*0.003*	*0.005*	*0.003*
Equals one if analyst initiates coverage	0.000	0.009	−0.001	0.009
	0.006	*0.006*	*0.006*	*0.006*
Log number of quarters since analyst initiated coverage (seasoning)	0.019***	0.021***	0.017***	0.020***
	0.002	*0.002*	*0.002*	*0.002*
Jobmove (analyst changing employer)	−0.040***	−0.061***	−0.041***	−0.061***
	0.011	*0.010*	*0.011*	*0.010*
Log number of stocks covered by analyst	−0.043***	−0.037***	−0.043***	−0.037***
	0.004	*0.003*	*0.004*	*0.003*
Company characteristics				
Log number of analysts covering stock	0.089***	0.080***	0.085***	0.077***
	0.004	*0.004*	*0.004*	*0.004*
Log company's dollar equity proceeds prior five years	−0.007***	−0.006***	−0.007***	−0.006***
	0.001	*0.001*	*0.001*	*0.001*
Log company's dollar debt proceeds prior five years	0.0019**	0.0016	0.0019**	0.0015
	0.0006	*0.0009*	*0.0006*	*0.0009*
Log company's dollar equity market capitalization	0.003*	0.002	−0.001	−0.003
	0.001	*0.002*	*0.001*	*0.002*
Random analyst effects	Yes	No	Yes	No
Random firm effects	No	Yes	No	Yes
Fraction of variance resulting from random effects (ρ) (percent)	14.7	3.3	14.8	3.3
Wald test: all coefficients = 0 (χ^2)	118.8***	146.3***	113.8***	143.1***
Overall GLS R^2 (percent)	1.2	1.4	1.2	1.4

Table 5, part 2

6.6 Table 6: Analyst forecast accuracy

Read:

- The dependent variable is relative forecast error, so negative coefficients imply greater accuracy.
- Equity underwriting relationships are associated with smaller forecast errors.

- Analysts at banks with larger brokerage operations are also more accurate.
- Institutional ownership has a negative coefficient, meaning forecast errors are smaller when institutional ownership is higher.
- Analysts with better past relative forecast accuracy continue to be more accurate.
- More senior analysts have larger errors in this specification.

Economic message: Institutional ownership improves not only recommendation discipline but also information quality. Analysts appear to work harder or be more careful when institutions are watching.

Table 6
Determinants of analyst forecast accuracy

The dependent variable is analyst forecast accuracy relative to consensus, defined as the absolute error of analyst *i*'s fiscal year-end forecast for company *k* in quarter *t* relative to company *k*'s subsequently realized earnings, scaled by the standard deviation of the forecasts made by all analysts covering company *k*'s stock in quarter *t*: *abs*(realized earnings – forecast earnings)/standard deviation. We then normalize this measure by subtracting the scaled absolute error of the contemporaneous consensus forecast (taken to be the median forecast) to ensure comparability across companies and across forecast horizons. We estimate two random-effects generalized least squares (GLS) models, including either analyst or firm random effects, respectively. To conserve space, intercepts are not shown. Standard errors are shown in italics underneath the coefficient estimates. We use ***, **, and * to denote significance at the 0.1%, 1%, and 5% level (two-sided), respectively. The number of observations is 221,521.

	Dependent variable: analyst forecast accuracy	
Independent variables	Random analyst effects (1)	Random firm effects (2)
Investment banking pressure (bank-company relationships)		
Bank's share of company's equity deals prior five years	−0.034***	−0.037***
	0.007	*0.007*
Bank's share of company's debt deals prior five years	0.000	−0.010
	0.010	*0.010*
Brokerage pressure (size of brokerage)		
Log number of registered representatives	−0.009***	−0.009***
	0.001	*0.001*
Bank characteristics		
Bank's loyalty index	−0.024	−0.021
	0.013	*0.012*
Bank's equity market share prior calendar year	−0.077	−0.090*
	0.046	*0.039*
Institutional ownership		
Percent institutional ownership	−0.018*	−0.021*
	0.009	*0.009*
Analyst characteristics		
Equals one if analyst is an *Institutional Investor* "all-star"	−0.001	−0.006
	0.005	*0.004*
Relative forecast accuracy	−0.767***	−0.819***
	0.020	*0.018*
Log analyst's seniority (in years)	0.011***	0.011***
	0.004	*0.003*
Equals one if analyst initiates coverage	0.007	0.009
	0.007	*0.007*
Log number of quarters since analyst initiated coverage (seasoning)	0.001	0.001
	0.002	*0.002*
Jobmove (analyst changing employer)	−0.019	−0.026*
	0.013	*0.013*
Log number of stocks covered by analyst	−0.009*	0.000
	0.004	*0.004*
Company characteristics		
Log number of analysts covering stock	0.018***	0.016***
	0.004	*0.004*
Log company's dollar equity proceeds prior five years	0.003***	0.003**
	0.001	*0.001*

Table 6, part 1

Table 6 (*continued*)

	Dependent variable: analyst forecast accuracy	
Independent variables	Random analyst effects (1)	Random firm effects (2)
Log company's dollar debt proceeds prior five years	−0.001	0.000
	0.001	*0.001*
Log company's dollar equity market capitalization	−0.005***	−0.005**
	0.001	*0.001*
Random analyst effects	Yes	No
Random firm effects	No	Yes
Fraction of variance resulting from random effects (ρ) (percent)	0.4	0
Wald test: all coefficients = 0 (F)	97.3***	135.4***
Overall GLS R^2 (percent)	1.0	1.0

Table 6, part 2

6.7 Table 7: Timeliness of reactions to bad news

Read:

- The dependent variable is log time to recommendation revision after a large negative price shock.
- Larger price drops lead to faster recommendation revisions.
- All-star analysts react faster in the 4X price-drop specifications.
- More senior analysts and analysts with longer coverage history react more slowly.
- Institutional ownership is negative and significant: analysts react faster to bad news when institutions own more of the stock.
- Investment banking relationship variables do not consistently delay reactions after controlling for other factors.

Economic message: Institutions discipline analyst timeliness. In stocks heavily owned by institutions, analysts are less willing to delay bad news.

Table 7
Timeliness of reactions to bad news

The sample consists of event-analyst pairs, in which the event is defined as a one-day split-adjusted share price drop that exceeds four or five times the company's prior-year standard deviation of daily returns. These are unusually large price drops. We require consecutive events for the same firm to be at least three months apart. For each such event, we find each instance in which an analyst (employed at one of our sample banks) covered the company's stock in the 365 days prior to the event (so there are multiple event-analyst pairs for each event). We then compute the time it takes the analyst to react to the price drop by issuing a new report and regress it on the magnitude of the event-day price drop and the variables introduced in Table 2. We restrict the estimation sample to reaction times of 365 days or less, to focus on recommendation revisions that are likely related to the event. Similar (albeit noisier) results obtain with longer windows, and if we use accelerated-time-to-failure (duration) models. All models are estimated using least squares and include analyst firm fixed effects. To conserve space, intercepts and fixed effects are not shown. Standard errors are shown in italics underneath the coefficient estimates. We use ***, **, and * to denote significance at the 0.1%, 1%, and 5% level (two-sided), respectively.

	Dependent variable: *ln* time to recommendation revisions			
	4X price drop		5X price drop	
Independent variables	(1)	(2)	(3)	(4)
Magnitude of price drop (one-day return)	3.469***	3.145***	3.021***	2.669***
	0.214	*0.203*	*0.284*	*0.267*
Analyst characteristics				
Equals one if analyst is an *Institutional Investor* all-star	−0.136*	−0.143*	−0.076	−0.086
	0.061	*0.061*	*0.089*	*0.089*
Relative forecast accuracy	−0.435	−0.427	−0.990*	−1.001*
	0.308	*0.309*	*0.459*	*0.460*
Log analyst's seniority (in years)	0.747***	0.721***	0.717***	0.682***
	0.073	*0.073*	*0.109*	*0.108*
Log number of quarters since analyst initiated coverage (seasoning)	0.300***	0.266***	0.326***	0.284***
	0.023	*0.022*	*0.033*	*0.032*
Jobmove (analyst changing employer)	0.054	0.048	0.108	0.098
	0.080	*0.080*	*0.122*	*0.123*

Table 7, part 1

Table 7 (*continued*)

	Dependent variable: *ln* time to recommendation revisions			
	4X price drop		5X price drop	
Independent variables	(1)	(2)	(3)	(4)
Log number of stocks covered by analyst	−0.223***	−0.216***	−0.369***	−0.361***
	0.048	*0.048*	*0.071*	*0.071*
Company characteristics				
Log number of analysts covering stock	−0.070		−0.064	
	0.040		*0.059*	
Log company's dollar equity proceeds prior five years	0.010		0.017	
	0.007		*0.010*	
Log company's dollar debt proceeds prior five years	−0.007		−0.018	
	0.007		*0.010*	
Log company's dollar equity market capitalization	−0.043**		−0.037	
	0.014		*0.021*	
Investment banking pressure (bank-company relationships)				
Bank's share of company's equity deals prior five years	0.033	0.119*	−0.007	0.096
	0.060	*0.057*	*0.086*	*0.083*
Bank's share of company's debt deals prior five years	−0.050	−0.080	−0.172	−0.240
	0.099	*0.097*	*0.148*	*0.146*
Bank characteristics				
Bank's loyalty index	−0.266*	−0.270*	−0.560**	−0.561**
	0.132	*0.132*	*0.197*	*0.197*
Bank's equity market share prior calendar year	−0.901	−0.894	−0.225	−0.233
	0.649	*0.650*	*0.927*	*0.928*
Institutional ownership				
Percent institutional ownership	−0.210*	−0.325***	−0.372**	−0.476***
	0.086	*0.083*	*0.126*	*0.122*
Analyst fixed effects	Yes	Yes	Yes	Yes
Wald test: all coefficients = 0 (F)	41.5***	71.6***	21.4***	21.4***
Adjusted R^2 (percent)	11.4	11.0	10.8	10.4
Number of observations	9,956	9,956	5,842	5,842

Table 7, part 2

7 Short summary

- The paper studies whether institutional investors moderate conflicts of interest in sell-side research.
- Analysts face pressure from investment banking relationships and brokerage commission incentives.
- Analysts also face career and reputation incentives created partly by institutional investors.
- The main sample covers 230,268 analyst-firm-quarter recommendations over 1994–2000.
- Investment banking relationships are associated with more favorable relative recommendations.
- Brokerage pressure is also associated with more favorable relative recommendations.
- Institutional ownership is associated with more conservative recommendations.
- Institutional ownership is associated with more accurate earnings forecasts.
- Institutional ownership is associated with faster recommendation revisions after severe stock price drops.
- Robustness tests address analyst and firm random effects, Fama-MacBeth estimation, ordered probit models, and endogenous coverage selection.
- The central lesson is that institutional investors can discipline not only corporate managers but also information intermediaries.

8 Conclusion

8.1 Core conclusion

Ljungqvist, Marston, Starks, Wei, and Yan show that sell-side analysts respond to multiple and competing incentives. Investment banking relationships and brokerage incentives push analysts toward more favorable recommendations. Institutional ownership pushes in the opposite direction by increasing the reputational cost of biased or low-quality research.

8.2 Economic intuition

Institutional investors are important consumers of sell-side research. They vote in analyst rankings and influence trading commission allocations. Analysts therefore have incentives to maintain credibility with institutions. When a stock is highly visible to institutions, biased research is more likely to be detected and punished. This disciplining force offsets conflicts created by investment banking and brokerage businesses.

8.3 Incremental contribution takeaway

The paper's incremental contribution is to identify the institutional investor as a moderating force in the analyst conflict-of-interest problem. It integrates three literatures–investment banking conflicts, brokerage incentives, and institutional investor monitoring–and shows that institutional investors affect recommendation optimism, forecast accuracy, and reaction speed to bad news.

8.4 Final takeaway

The paper does not claim that conflicts disappear. Instead, it shows that conflicts are context-dependent. Analysts are more biased when institutional scrutiny is weak and more disciplined when institutional investors are present. The quality of financial intermediation therefore depends on the monitoring environment created by sophisticated investors.